

“CityPulse” is a startup working with local councils to improve urban transport. The company collects data from multiple sources:

- GPS traces from **bus fleets** (every few seconds)
- **Ticketing transactions** (batch uploaded nightly)
- **Citizen feedback reports** from a web/mobile platform (stored as documents)
- **External APIs** providing weather and events (streaming updates)

The system must:

- Support *millions of daily records* from IoT and mobile devices.
- Provide **real-time congestion alerts**.
- Train **predictive models** to forecast travel demand by zone.
- Serve **interactive dashboards** for city planners.

Initially, CityPulse used a single PostgreSQL database, but as data volumes and latency requirements grew, it began to fail under load. The team decided to **migrate to a distributed architecture**.

Component	Old System	New System
Data storage	Central PostgreSQL instance	Distributed NoSQL cluster (Cassandra / MongoDB)
Data format	Relational tables	Document-based records (JSON-like structure per bus, route, or feedback)
Model training	Single server, full dataset batch training	Mini-batch & online learning with streaming data
Consistency	Strict ACID	Eventual consistency
Replication	Manual nightly backup	Multi-node replication across regions
Data access pattern	Fixed queries	Ad hoc analytics and dashboards
Historical data storage	On-disk tables	SSTables for efficient writes and compactions